25900669 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course(VAC) - MG3VACTXLA01 - CHEMICAL PROCESSING IN TEXTILES

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions
Answer all questions
Each question carries **1** marks

1. Under which classification does flame-resistant finish come? [U]/[CO1]
   Ans: Functional finish
2. What is a disadvantage of stock dyeing? [U] / [CO3]
   Ans: Expensive and time-consuming
3. What type of dye is commonly used for dyeing wool and silk fibres? [K]/[CO3]
   Ans: Acid dyes
4. In direct printing, how is the design applied to the fabric? [U]/[CO4]
   Ans: By applying printing paste directly on fabric
5. What component is not required in the jet printing process? [U]/[CO4]
   Ans: All of the above
6. What is the primary benefit of using nanotechnology in textile finishing? [U] / [CO2]
   Ans: All of the above
7. In flat-bed screen printing, the number of screens corresponds to: [U]/[CO4]
   Ans: Number of colors
8. What happens if fabric is over bleached? [K] / [CO2]
   Ans: Fiber damage and yellowing
9. Which of the following finishes is considered temporary and removed after washing? [K] / [CO2]
   Ans: simple calendering
10. What is the purpose of reducing agents in vat dyeing? [U] / [CO3]
    Ans: To increase dye solubility

[1x10 = 10]

Part B

Fill in the Blanks Questions

Answer all questions

Each question carries 1 marks

11. Gigging was traditionally done using brushes made of _____ plants. [U] / [CO2]
    Ans: Teasel
12. Solution dyeing involves adding colorants to the _____solution [U] / [CO3]
    before fibre production.
    Ans: spinning (or dope)
13. Basic or routine finishes are also known as_____finishes. [K]/[CO1]
    Ans: Preparatory
14. At the manufacturing level, moth-proofing can involve modifying the sulphur-[K] / [CO2]
    containing _____ in wool.
    Ans: amino acids
15. In product dyeing garments are dyed after _____. [U] / [CO3]

    Ans: sewing (or assembly/construction)

16. To ensure proper alignment when stamping, many blocks have _____ [U] / [CO4]
    carved into them.
    Ans: Pitch points
17. Acid dyes are suitable for dyeing _____fibers such as nylon [U] / [CO3]

    Ans: synthetic fibres

18. Stencil printing influenced the later development of _____ printing. [U] / [CO4]

    Ans: Screen printing

19. Discharge printing is most effective on fabrics dyed with _____ dyes. [U] / [CO4]
    Ans: Dark (or deep)

20. Soil and stain-resistant finishes reduce the need for frequent washing or heavy [U] / [CO2]
_____.
    Ans: Detergents

[1x10 = 10]

Part C

Answer in One Word or One Sentence

Answer all questions

Each question carries 1 marks

21. Which natural method helps reduce microbes on fabrics at home? [K] / [CO2]

    Ans: Sunlight

22. What controls the design in digital textile printing? [U] / [CO4]

Ans: Digital computer

23. What is union dyeing? [U] / [CO3]

Ans: Union dyeing is a dyeing process used for fabrics made of blended fibers (e.g., a cotton/polyester blend) that aims to color all the different fibers the same solid shade using a combination of different dye classes in a single bath.

24. What is the common application of plasma technology? [K] / [CO2]

Ans: The common application is surface modification to make fabrics more absorbent, water-repellent, flame-retardant, antimicrobial, or better at dye uptake and adhesion of coatings.

25. Under which classification does sueding come? [K] / [CO1]

Ans: Aesthetic finish/Mechanical finish

[K] / [CO2]

26. Which technique is used to deposit fibers in flocking besides mechanical methods?

Ans: Electrostatic (or electrostatically)

27. What is dope dyeing? [U] / [CO3]

Ans: Dope dyeing is when color is added to the liquid material (the dope or polymer solution) before it is spun into a synthetic fiber.

28. What is a mordant? [U] / [CO4]

Ans: A mordant is a substance, usually a metallic salt, that acts as a chemical bridge to help a dye bond permanently with a fiber, thus improving colorfastness.

29. What is a characteristic of reactive dyes? [K] / [CO3]

Ans: The primary characteristic of reactive dyes is that they form a strong chemical bond directly with the fiber molecules, resulting in color that is highly resistant to washing and fading.

30. Name one other application for sublimation methods besides textiles.

Ans: home furnishings, promotional items, Ceramics [U] / [CO4]

[1x10 = 10]

Part D

Short Answer Questions

Answer any 5 questions

Each question carries 2 marks

31. In functional finishes, explain how water-repellent finishes improve the comfort and utility of fabrics. [K] / [CO1]

Ans: Water-repellent finishes improve comfort by preventing the fabric from getting soaked (keeping the wearer dry) while still allowing air to pass through (making it breathable). They improve utility by making the fabric resist getting wet so it can be used for items like outdoor or sports clothing that need to stay dry for long periods.

32. Why is shrinkage resistant finishing important for wool fabrics? [U]/ [CO2]

Ans: Shrinkage resistant finishing is important for wool fabrics because they can suffer from felting shrinkage, which occurs when the scales on wool fibers interlock due to moisture, heat, and movement during washing, leading to a permanent reduction in size.

33. What is an advantage of yarn dyeing? [K] / [CO3]

Ans: The main advantage of yarn dyeing is that it allows for the creation of intricate, multicolored patterns (like plaids, stripes, checks, or Ikat) that are woven directly into the fabric. This method ensures deep, uniform color penetration within the yarns, which results in patterns that are more durable and colorfast compared to designs simply printed onto the finished fabric surface.

34. Why is sublimation printing not suitable for untreated cotton? [U]/ [CO4]

Ans: Sublimation printing is not suitable for untreated cotton because the high heat causes the special dyes to turn into a gas, and this gaseous dye can only bond effectively with synthetic fibers like polyester, meaning the color will not stick permanently to cotton.

35. Explain the embossing process in textile finishing? [U]/ [CO2]

Ans: The embossing process involves passing the fabric between two rollers: a metal roller with a raised design and a soft roller (made of cloth or paper). The metal roller presses the design into the soft roller, and when the fabric passes between them, it picks up the raised design, creating a visible, three-dimensional texture and pattern on its surface.

36. What is the main advantage of jet dyeing? [U]/ [CO3]

Ans: The main advantage of jet printing is that it is a non-contact, computer-controlled method that requires no screens, blocks, or rollers, enabling the production of highly detailed, complex, and customized designs in one fast step, while also being cost-effective for small production runs. [2x5 = 10]

Part E

Long Answer Questions

Answer any 2 questions

Each question carries 5 marks

37. Explain the process and importance of durable press finishes in textiles. [U]/ [CO2]

Ans: Durable press finishes make clothes resist wrinkles after washing and wearing, eliminating the need for constant ironing.

How it Works

1. Chemicals (resins) are applied to the fabric, primarily to cotton.
2. These resins form strong, lasting chemical bonds inside the fiber.
3. The fabric or garment is then "baked" at high heat in a process called curing.
4. This curing step locks the fabric's smooth shape in place, so the material returns to that smooth state even after it's crumpled or washed.

38. Explain aesthetic finishes and their two types? [U]/ [CO2]

Ans: Aesthetic finishes are treatments applied to a fabric to primarily change its appearance and/or hand (feel/texture).

1. Special Calendering (Luster/Pattern Finishes): These finishes primarily change the way light reflects off the fabric's surface, adding shine, gloss, or a specific visual pattern.
    - Examples include Schreinerizing (adds subtle luster) and Moiré (creates a wavy, water-like pattern on ribbed fabrics).
    - Embossing also falls here by pressing a three-dimensional pattern into the surface.
2. Raised Surface Finishes (Texture Finishes): These finishes involve mechanical techniques that lift or raise the fiber ends from the surface of the fabric, improving texture and feel.
    - Examples include Napping (creates a fuzzy layer for softness/warmth), Gigging (gently raises fibers using teasels), Sueding (creates a soft, velvety texture using abrasive rollers), and Flocking (depositing short fibers with adhesive).

39. What is dip dyeing, and why are fabrics dipped repeatedly? [U]/ [CO3]

Ans: What is dip dyeing? Dip dyeing is a manual technique where fabric, yarn, or garments are dipped into a dye bath either partially or completely to create gradient, ombré, or two-tone effects.

Why are fabrics dipped repeatedly? Fabrics are dipped repeatedly to control the intensity and coverage of the colour over different sections of the item, which is necessary to achieve the desired gradient or ombré effect.

40. Explain the roller printing process step by step, including the main machinery and components involved. [U]/[CO4]

Ans: Step-by-Step Process

1. Engraving: Large copper rollers are etched with the pattern and coated with chromium. The size of the design determines the roller diameter.
2. Color Pickup: Each roller picks up color from a dye trough.
3. Color Control: A doctor blade wipes off excess paste, ensuring color remains only in the engraved parts (the grooves).
4. Transfer: The fabric is passed between the rollers, and the printing paste held in the grooves is then transferred to the fabric, creating the pattern.
5. Pattern Creation: The process is repeated simultaneously for each color, with each roller printing its part of the design to create a sharp and continuous multicolor pattern.

[5x2 = 10]